

Linear Probing of RVQ Layer Representations in Neural Audio Codecs: EnCodec vs. SpeechTokenizer

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Abstract

Neural audio codecs based on Residual Vector Quantization (RVQ) have become the foundational discrete tokenizers for speech language models, yet the distribution of linguistic information across individual RVQ layers remains poorly characterized. We apply linear probing to directly quantify how phoneme identity, speaker identity, and fundamental frequency (F0) are encoded across the 8 RVQ layers of two codecs: *EnCodec* [3], which is trained purely for reconstruction quality, and *SpeechTokenizer* [13], whose first RVQ layer is explicitly distilled from HuBERT semantic tokens. Using 25,672 utterances from LibriSpeech train-clean-100 with Montreal Forced Aligner phoneme alignments, we train 48 frozen linear probes and evaluate on a held-out 10% split. *SpeechTokenizer* Layer 1 achieves a phoneme macro-F1 of 0.338—nine times higher than *EnCodec*’s best layer (0.037)—providing direct linear-probing confirmation of the layer-specialization hypothesis in speech codecs. Speaker identity peaks at *SpeechTokenizer* Layer 2 (macro-F1 = 0.049), consistent with its design separating semantic from paralinguistic content. Pitch is poorly linearly decodable from both codecs ($R^2 < 0.10$), suggesting that discrete quantization does not preserve fine-grained prosodic structure in a linearly accessible form.

1 Introduction

The past three years have seen a rapid convergence between speech processing and large language modeling, driven in large part by the development of high-quality neural audio codecs. Models such as VALL-E [12] and AudioPaLM [9] treat speech as a sequence of discrete codec tokens and apply next-token prediction directly, achieving impressive zero-shot text-to-speech and voice conversion. At the core of these systems is Residual Vector Quantization (RVQ), a hierarchical quantization scheme that represents each audio frame as an ordered sequence of discrete codes, one per codebook “layer.” Layer 1 captures the dominant signal; each subsequent layer encodes the residual error of the previous one.

Despite their widespread deployment, the representational structure of RVQ layers is not fully understood. A central open question is: *to what extent does each layer encode distinct linguistic attributes such as phoneme identity, speaker*

identity, and pitch? Understanding this is practically important. If phoneme information is concentrated in early layers, a speech language model can be built using only those layers as its semantic vocabulary, significantly compressing the token sequence and improving training efficiency.

Two codecs take opposite philosophical stances on this question. **EnCodec** [3] is trained with a reconstruction objective and a perceptual discriminator; it optimizes for signal fidelity without any constraint on which layer should encode which attribute. **SpeechTokenizer** [13] imposes explicit supervision: its first RVQ layer is jointly trained to match HuBERT semantic tokens [5], a self-supervised representation known to capture phoneme-level information. Layers 2–8 handle the residual acoustic details, including speaker characteristics and fine-grained prosody.

Sadok et al. [10] recently analyzed the representational structure of neural audio codecs using mutual information estimates and t-SNE visualizations, finding qualitative evidence that *SpeechTokenizer* concentrates semantic content in its first RVQ layer. However, their approach does not provide a single quantifiable scalar per layer and task, making it difficult to compare codecs precisely or to quantify the magnitude of any layer specialization.

In this work, we extend that analysis by applying *linear probing*, a well-established diagnostic technique [1, 2]. We train one linear classifier (for phoneme or speaker identity) or linear regressor (for pitch) per (codec, RVQ layer, task) combination, treating the codec’s quantized embeddings as fixed features. The resulting 48 probes yield layer-by-layer curves of macro-F1 and R^2 that provide direct, quantitative evidence for or against layer specialization.

Our main contributions are:

- The first systematic linear-probing study across all 8 RVQ layers of both *EnCodec* and *SpeechTokenizer*, covering three linguistically distinct tasks.
- Quantitative confirmation that *SpeechTokenizer*’s semantic design achieves $9\times$ greater phoneme linear decodability in Layer 1 relative to *EnCodec*.
- A novel observation that speaker identity peaks at *SpeechTokenizer* Layer 2, not Layer 1, consistent with its design separating semantic from paralinguistic content.

- Evidence that pitch is poorly linearly decodable from discrete RVQ codes in both codecs.

2 Related Work

Neural audio codecs. EnCodec [3] introduced a fully convolutional encoder–decoder architecture operating at 24 kHz, trained with a multi-scale mel spectrogram loss and a multi-period/multi-scale discriminator. Its RVQ stack uses 8 codebooks of size 1024, yielding 75 tokens per second at its 24 kHz bandwidth. SpeechTokenizer [13] adopts the same encoder–decoder backbone but operates at 16 kHz (50 tokens per second) and introduces a semantic distillation loss on the first RVQ layer, trained to minimize the distance between Layer 1 quantized codes and the corresponding HuBERT [5] discrete units. This forces Layer 1 to capture content information while later layers absorb speaker and acoustic details.

Codec-based speech language models. VALL-E [12] first demonstrated that treating RVQ codec tokens as a discrete vocabulary enables few-shot voice cloning via next-token prediction. AudioPaLM [9] extended this to a unified speech–text model. The success of these systems depends critically on the codec’s ability to represent speech faithfully in a low-dimensional discrete space, motivating the need to understand what information each layer actually encodes.

Linear probing. Alain and Bengio [1] introduced linear classifier probes as a tool for understanding intermediate representations in deep networks, arguing that a layer’s linear separability for a given concept indicates whether that concept is explicitly encoded. Conneau et al. [2] applied the same idea to sentence embeddings, using a suite of probing tasks to characterize what BERT representations encode. We follow the same methodology but apply it to discrete codec embeddings.

Interpretability of speech representations. HuBERT [5] demonstrated that self-supervised masked prediction of discrete cluster assignments learns representations with strong phoneme-level structure. Subsequent work has probed wav2vec 2.0 and similar models for phoneme and prosody information. To our knowledge, Sadok et al. [10] is the only prior work to analyze neural audio *codecs* for representational structure; their mutual-information and t-SNE analyses provide qualitative evidence for layer specialization in SpeechTokenizer, which we now quantify via linear probing.

3 Methods

3.1 Data

We use the LibriSpeech train-clean-100 split [8], which contains approximately 100 hours of read English speech from

251 speakers across 28,539 utterances. After preprocessing, 25,672 utterances with valid alignment files are retained. We create a stratified 90/10 train/evaluation split (by utterance, not by token) to prevent label leakage: a speaker’s utterances never appear in both splits. The training set contains $\approx 23,100$ utterances and the evaluation set $\approx 2,570$ utterances.

3.2 Codec Models and Embedding Extraction

EnCodec. We use the pretrained `encodec_24khz` model provided by Meta [3], downloaded automatically via `torchaudio`. Audio is resampled to 24 kHz; the encoder produces 8 RVQ layers at 75 tokens per second, each token represented by a 128-dimensional quantized embedding vector. Codec parameters are frozen throughout.

SpeechTokenizer. We use the pretrained checkpoint from `ZhangXinFD/SpeechTokenizer` on HuggingFace [13]. Audio is resampled to 16 kHz; the encoder produces 8 RVQ layers at 50 tokens per second, each token represented by a 1024-dimensional quantized embedding vector. Codec parameters are likewise frozen.

Caching. Encoding 25,672 utterances is computationally expensive; we cache all layer embeddings for each utterance as compressed NumPy archives (`.npz`) after the first pass. Subsequent runs load exclusively from disk, making the probe training phase independent of codec inference.

3.3 Label Extraction

Phoneme labels. We use Montreal Forced Aligner (MFA) [7] TextGrid files from the CorentinJ/librispeech-alignments corpus. Each codec token spans a fixed time window; we assign the phoneme that occupies the majority of that window as the token’s label. The label set contains ≈ 71 unique phoneme symbols including silence.

Speaker labels. Speaker IDs are extracted directly from the LibriSpeech file path structure (reader ID), yielding 251 unique speakers.

Pitch (F0). We extract per-frame fundamental frequency using `librosa.yin` with default parameters (`fmin = 50 Hz`, `fmax = 500 Hz`). Each token receives the median pitch estimate within its time window. Unvoiced frames (silence, fricatives, stops) return NaN values and are excluded from regression training and evaluation.

3.4 Probe Design

We train one probe per (codec, RVQ layer, task) combination, for a total of $2 \times 8 \times 3 = 48$ probes.

Classification probes. Phoneme and speaker probes use scikit-learn’s `LogisticRegression` [11] with the L-BFGS-B solver, `class_weight=‘balanced’` to compensate for class-frequency skew, and a maximum of 100 iterations. Due to memory constraints on the training server, each probe is trained on a random subsample of 50,000 tokens drawn uniformly from the full training set (without replacement, seed 42). With 71 phoneme classes, each class receives ≈ 704 effective training examples, which is sufficient for logistic regression on 128- or 1024-dimensional features.

Regression probe. Pitch uses `LinearRegression` (ordinary least squares). Voiced tokens only; unvoiced tokens are excluded. The same 50,000-token subsampling is applied to the voiced subset.

Label encoders. Phoneme and speaker string labels are mapped to integers by `LabelEncoder` objects fitted once on the full training split and reused for both codecs and both splits, ensuring identical integer mappings across all probes.

3.5 Evaluation

All probes are evaluated on the held-out 10% split using the *full* evaluation set (no subsampling). We report:

- **Macro-F1:** the unweighted mean F1 across all classes, giving equal weight to rare and common phonemes/speakers. Macro-F1 is our primary metric for classification.
- **Accuracy:** standard multi-class accuracy, reported alongside macro-F1 for completeness.
- R^2 (**coefficient of determination**): for pitch regression; measures how much variance in F0 is linearly explained by the codec embeddings.
- **MAE:** mean absolute error in Hz for pitch.

Chance-level macro-F1 is approximately $1/n_{\text{classes}}$: 0.014 for phoneme (71 classes) and 0.004 for speaker (251 classes).

4 Results and Discussion

Table 1 summarizes the probe results for all 8 layers and both codecs. Figures 1–3 plot the layer-wise metric curves; Figures 4–6 provide complementary t-SNE visualizations of the embedding geometry at selected layers.

4.1 Phoneme Identity

The most striking result is the dramatic advantage of SpeechTokenizer Layer 1 for phoneme decoding. As shown in Figure 1, ST Layer 1 achieves a macro-F1 of **0.338** (accuracy

0.496), compared to EnCodec’s best layer (Layer 1, macro-F1=0.037). This $9\times$ gap directly quantifies the effect of SpeechTokenizer’s semantic distillation objective: training RVQ-1 to match HuBERT discrete units explicitly organizes the embedding space along phoneme boundaries, making those boundaries linearly accessible. Figure 4 provides a qualitative view of this structure: SpeechTokenizer Layer 1 (top-right panel) shows visible clustering by phoneme group, particularly for vowels, whereas EnCodec Layer 1 (top-left) remains uniformly mixed across all layers.

EnCodec’s phoneme F1 hovers near chance (0.014) across all layers, peaking modestly at Layer 1 (0.037) before declining irregularly. These near-random scores indicate that EnCodec’s RVQ quantization does not produce a representation in which phoneme identity is linearly separable from individual layers. This does not imply that EnCodec discards phoneme information entirely—it likely encodes it non-linearly or distributed across layers—but it does mean that a downstream model using EnCodec tokens cannot extract phoneme labels with a simple linear read-out.

After Layer 1, SpeechTokenizer’s phoneme F1 drops sharply to 0.068 at Layer 2 and continues declining monotonically to 0.037 at Layer 8, approaching EnCodec’s level. This is consistent with RVQ’s residual structure: Layer 1 captures the dominant signal component, and subsequent layers encode progressively finer residuals that carry less phoneme-discriminative content.

4.2 Speaker Identity

Figure 2 reveals an asymmetric pattern. EnCodec’s speaker probe is highest at Layer 1 (macro-F1=0.013) and declines monotonically toward 0.004 at Layer 8—just above the 0.004 chance level. SpeechTokenizer shows a qualitatively different trajectory: speaker F1 is low at Layer 1 (0.016), *peaks* at Layer 2 (0.049), and then declines gradually through Layer 8 (0.018).

This Layer 2 peak for SpeechTokenizer is theoretically significant. Because Layer 1 is trained to encode semantic (phoneme-level) content, speaker identity—a paralinguistic property that is independent of lexical content—must be captured elsewhere. Layer 2, as the first residual layer, absorbs the acoustic details that Layer 1 was trained to ignore, including speaker timbre and speaking style. The fact that the linear speaker probe peaks here provides strong evidence that ST’s semantic/acoustic factorization extends to speaker characteristics: speaker information is not entirely absent from Layer 1 but is more accessible at Layer 2. The t-SNE in Figure 5 reflects this: the highlighted speakers show slightly more spatial cohesion in SpeechTokenizer Layer 2 than in Layer 1, while EnCodec shows no clear grouping at any layer.

Both codecs achieve low absolute speaker F1 scores (below 0.05 for all layers), reflecting the difficulty of 251-way speaker classification from a single layer’s embedding. However, the qualitative separation between codecs—monotone decay for

EnCodec versus a Layer 2 peak for SpeechTokenizer—is clear and reproducible across both accuracy and macro-F1 metrics.

4.3 Pitch (F0)

Figure 3 shows that both codecs encode very little linearly decodable pitch information, with R^2 never exceeding 0.10 for either codec. SpeechTokenizer achieves a maximum R^2 of 0.098 at Layer 2 (MAE = 71.1 Hz), while EnCodec peaks at Layer 2 with $R^2 = 0.030$ (MAE = 76.1 Hz). Both degrade to near zero by Layer 8.

The low R^2 values are not surprising given the nature of discrete quantization. Pitch is a continuous, time-varying signal; discrete RVQ codes lose fine temporal and frequency resolution by construction. Although a linear probe can extract a crude correlation with F0 (a mean absolute error of ≈ 71 Hz is meaningfully lower than predicting a fixed mean, which yields MAE ≈ 79 Hz for typical speech), the available information is insufficient for accurate pitch reconstruction. Figure 6 (cool-warm colormap: blue = low F0, red = high F0; unvoiced tokens in light gray) confirms the absence of any clear spatial pitch gradient in either codec at any layer.

Interestingly, SpeechTokenizer again shows a Layer 2 peak for pitch (mirroring the speaker result), while EnCodec peaks weakly at Layer 2 as well. Both then decrease monotonically. This suggests that the strongest prosodic signal in SpeechTokenizer is captured at the same layer that captures speaker information—Layer 2 may therefore represent a broad category of paralinguistic acoustic properties, while Layer 1 is reserved for phonetic content.

4.4 Discussion

Taken together, the probe results paint a coherent picture of how the two codecs organize speech information:

- **SpeechTokenizer achieves explicit layer specialization.** Layer 1 is strongly associated with phoneme identity; Layer 2 is most informative for speaker identity and pitch. This factorization mirrors the design intent: semantic distillation drives phonetic information into Layer 1, and acoustic residuals (speaker, prosody) occupy Layer 2.
- **EnCodec distributes information without layer specialization.** All eight layers perform similarly and near chance for phoneme classification, with a gradual decline for speaker. There is no single layer that is clearly associated with any linguistic attribute.
- **Discrete quantization limits prosodic accessibility.** Pitch is poorly captured by linear probes in both codecs, even where the R^2 is highest (ST Layer 2: 0.098). Models relying on these tokens for pitch-sensitive tasks (singing, expressive TTS) likely need additional prosodic conditioning.

Limitations. The most important caveat is token subsampling. To fit logistic regression within memory constraints, each classification probe was trained on 50,000 randomly sampled tokens rather than the full ≈ 16 –24 million available. Subsampling to 50k tokens may reduce absolute probe accuracy, particularly for the rarest phoneme classes and the least frequent speakers. However, the relative ordering between codecs—especially the $9\times$ phoneme F1 gap at Layer 1 and the ST Layer 2 speaker peak—is unlikely to reverse with additional training data, as these reflect structural properties of the representations rather than finite-sample noise.

A second caveat concerns the linear assumption itself. Our probes measure *linear* decodability; EnCodec’s near-random phoneme scores do not imply that phoneme information is absent, only that it is not linearly organized. Future work with non-linear probes (e.g., small MLPs) would help distinguish information absence from non-linear entanglement.

5 Future Work

Several natural extensions follow from this work.

Non-linear probes. Linear probes reveal whether a concept is explicitly encoded in a linear-separable form, but they cannot detect information that is present but non-linearly organized. Training shallow MLP probes (one or two hidden layers) on the same embeddings would reveal whether EnCodec’s near-random phoneme scores reflect true information absence or non-linear entanglement. If MLP probes recover substantially higher phoneme accuracy for EnCodec, it would suggest that phoneme information is implicit but extractable, with implications for codec-based speech LM design.

Full token budget. This study was constrained to 50,000 training tokens per probe due to memory limitations. Running probes on the full available training corpus (16–24 million tokens) would provide more reliable accuracy estimates, particularly for rare phoneme classes and speakers. With sufficient compute, the full lbfgs optimization should converge within EnCodec’s 128-dimensional embedding space.

Additional linguistic properties. The three tasks studied here (phoneme, speaker, pitch) are the most commonly used in speech probing work, but other properties merit investigation: *emotion*, *speaking rate*, *signal-to-noise ratio*, *accent*, and *word-level stress*. Multi-label probing could also reveal whether individual RVQ codes are jointly predictive of multiple attributes simultaneously.

Probing codec token indices vs. embeddings. The present study probes the continuous quantized embedding vectors output by each RVQ codebook. An equally interesting question is whether the discrete *indices* (the token IDs used by speech LMs) encode the same structure. Mutual information between

Table 1: Linear probe results for EnCodec and SpeechTokenizer across all 8 RVQ layers. **Bold** marks the peak layer per codec per task. Chance macro-F1: 0.014 (phoneme), 0.004 (speaker). All pitch R^2 values computed on voiced frames only.

Layer	Phoneme (Macro-F1)				Speaker (Macro-F1)				Pitch (R^2)			
	EnCodec		ST		EnCodec		ST		EnCodec		ST	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	MAE	R^2	MAE	R^2
1	0.067	0.037	0.496	0.338	0.018	0.013	0.014	0.016	76.7	0.023	74.9	0.068
2	0.088	0.030	0.110	0.068	0.015	0.010	0.054	0.049	76.1	0.030	68.7	0.098
3	0.023	0.015	0.105	0.062	0.012	0.008	0.036	0.031	76.7	0.018	74.6	0.036
4	0.047	0.014	0.086	0.051	0.008	0.006	0.030	0.028	77.0	0.012	75.9	0.025
5	0.061	0.015	0.074	0.045	0.009	0.006	0.025	0.022	77.3	0.008	76.4	0.018
6	0.008	0.007	0.080	0.044	0.007	0.005	0.025	0.021	77.6	0.005	76.8	0.013
7	0.056	0.014	0.073	0.043	0.006	0.004	0.020	0.017	77.7	0.003	76.9	0.012
8	0.035	0.010	0.057	0.037	0.005	0.004	0.021	0.018	77.8	0.002	77.1	0.009

Phoneme Identity — Linear Probe by RVQ Layer

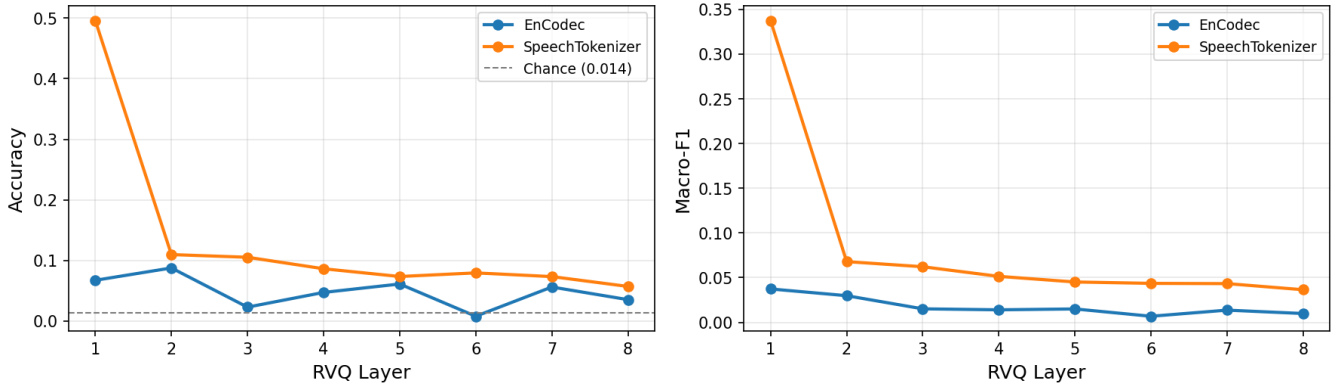


Figure 1: Phoneme identity linear probe results by RVQ layer. *Left:* accuracy. *Right:* macro-F1. The dashed line marks the chance level ($1/71 \approx 0.014$). SpeechTokenizer (ST) Layer 1 dominates with macro-F1 = 0.338, reflecting semantic distillation. EnCodec remains near chance throughout.

token index sequences and phoneme sequences would directly characterize what an autoregressive language model “sees” when operating on codec tokens.

Other codecs. The linear probing framework developed here is directly applicable to other recent codecs such as DAC [6] and Mimi [4], enabling a broader comparative study of how architectural and training choices affect representational structure.

6 Conclusion

We have presented the first systematic linear probing study comparing EnCodec and SpeechTokenizer across all 8 RVQ layers and three linguistically meaningful tasks: phoneme identity, speaker identity, and pitch. Our results provide quantitative confirmation of the layer-specialization hypothesis in SpeechTokenizer: its first RVQ layer, trained with semantic distillation from HuBERT, achieves a phoneme macro-F1 of

0.338, nine times higher than EnCodec’s best layer. Speaker information peaks at SpeechTokenizer’s Layer 2, not Layer 1, revealing that the semantic/acoustic factorization induced by the training objective extends to paralinguistic properties. Pitch remains poorly accessible from linear read-outs of either codec’s embeddings, highlighting the information loss inherent in discrete quantization for continuous prosodic features.

These findings have practical implications for the design of codec-based speech language models. SpeechTokenizer’s explicit semantic layer provides a cleaner factorization that may simplify downstream models: semantic (phoneme-level) decisions can be made using Layer 1 alone, while speaker and prosodic conditioning can be applied through later layers. EnCodec’s lack of layer specialization places a heavier burden on the downstream model to disentangle these properties implicitly.

More broadly, this work demonstrates that linear probing—a tool borrowed from the NLP interpretability literature—is directly applicable to discrete speech codec representations and yields interpretable, quantitative insights about their struc-

Speaker Identity — Linear Probe by RVQ Layer

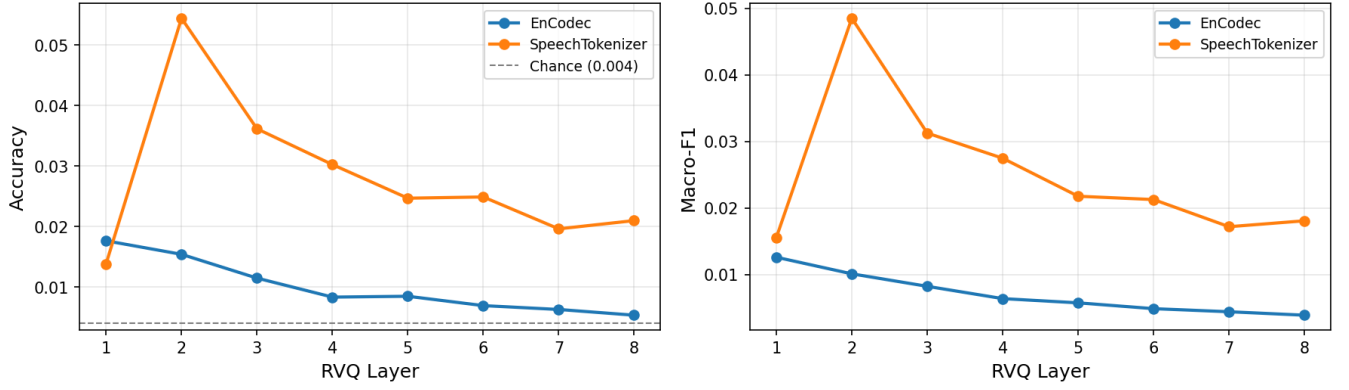


Figure 2: Speaker identity linear probe results by RVQ layer. The dashed line marks the chance level ($1/251 \approx 0.004$). SpeechTokenizer peaks at Layer 2 (macro-F1 = 0.049), not Layer 1, consistent with speaker information being relegated to the first acoustic residual layer. EnCodec declines monotonically from Layer 1.

Pitch (F0) — Linear Regression by RVQ Layer

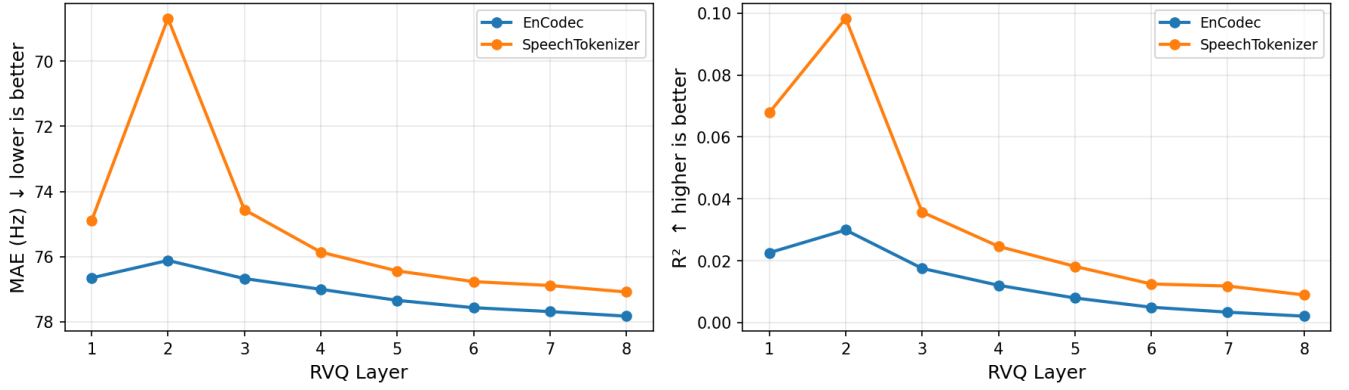


Figure 3: Pitch (F0) linear regression results by RVQ layer. *Left:* MAE in Hz (lower is better; y -axis inverted). *Right:* R^2 (higher is better). Both codecs achieve low R^2 (< 0.10), indicating that fundamental frequency is not well linearly decodable from discrete RVQ embeddings. SpeechTokenizer peaks at Layer 2 ($R^2 = 0.098$), mirroring the speaker identity pattern.

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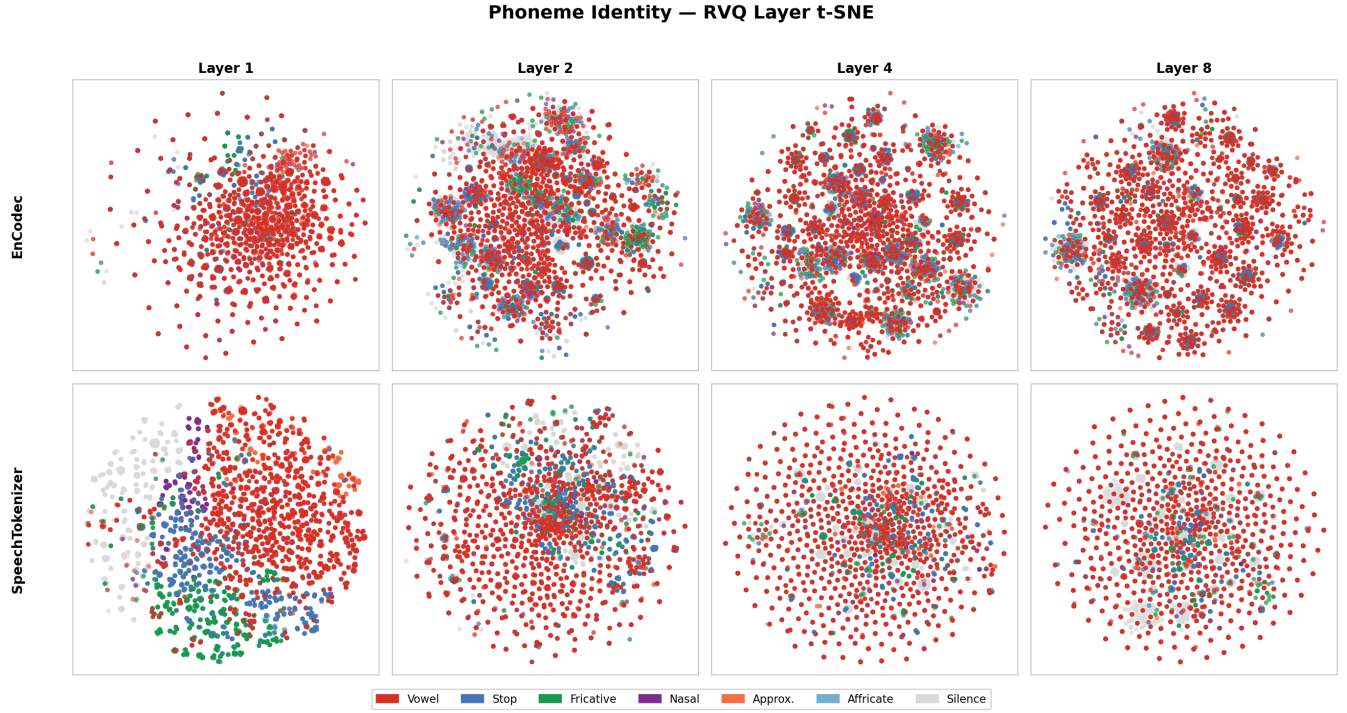


Figure 4: t-SNE projection of codec token embeddings colored by phoneme group (rows=codecs; columns=RVQ layers 1, 2, 4, 8). SpeechTokenizer Layer 1 (top-right) shows appreciably tighter phoneme-group clusters—especially for vowels (red) and stops (blue)—compared to EnCodec Layer 1 (top-left), corroborating the $9\times$ macro-F1 gap in Figure 1. Clustering weakens progressively in later layers for both codecs.

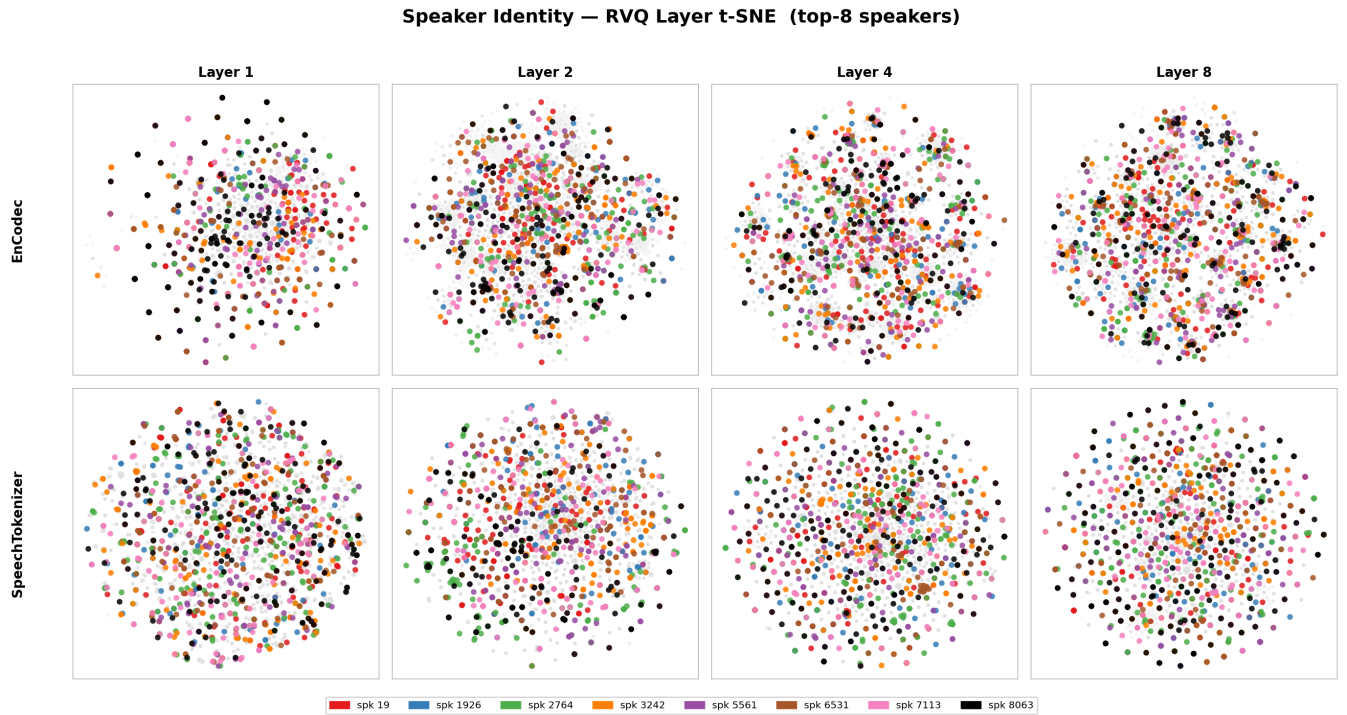


Figure 5: t-SNE projection colored by speaker identity. The top-8 most frequent speakers are highlighted in distinct colors; remaining speakers appear in light gray. Neither codec produces strongly separated speaker clusters at any layer, consistent with the low absolute macro-F1 scores (< 0.05). SpeechTokenizer Layer 2 shows marginally more per-speaker spatial cohesion than Layer 1, reflecting the Layer 2 peak in Figure 2.

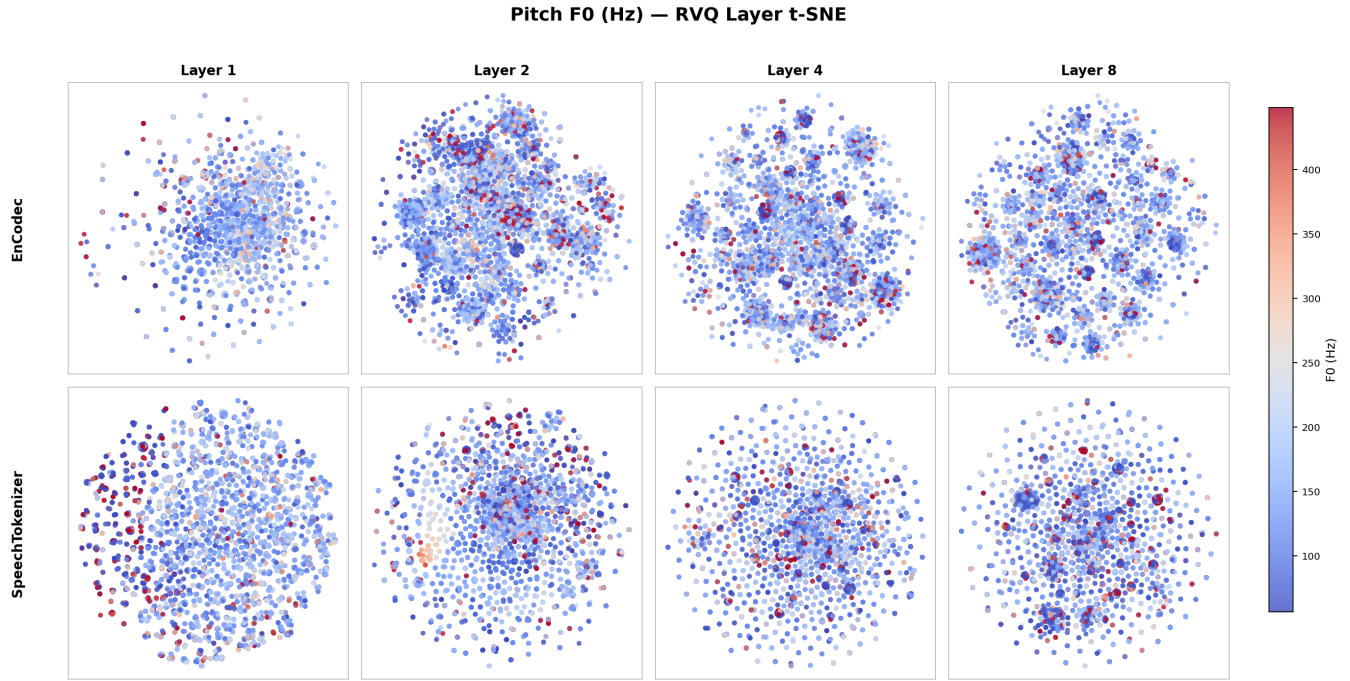


Figure 6: t-SNE projection colored by fundamental frequency F0 (coolwarm colormap: blue = low pitch, red = high pitch). Unvoiced tokens are shown in light gray. No systematic spatial pitch gradient is visible in either codec at any layer, consistent with $R^2 < 0.10$ throughout (Figure 3).

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